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Ecological factors influencing nest survival of hazel grouse *Bonasa bonasia* in a temperate forest, South Korea

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ABSTRACT

The survival of 86 clutches laid by hazel grouse *Bonasa bonasia* hens was studied using radio telemetry to evaluate the importance of key vegetation variables for nest survival between March 2003 and June 2009 in a 3000 ha study area of temperate forest in South Korea. The nest sites in the forest stands differed from that in the study area with dense understory cover and low visibility. Daily nest survival rates increased with understory cover, higher proportion of natural deciduous forest, and decreased with nest conspicuousness. Nest survival rates were not correlated with tree density, basal area, mid-story cover, and overstory cover. The results did not reveal any differences in the daily nest survival rate between first breeders and older hens. Habitat management that protects the remaining understory cover and natural deciduous forest is key to stabilizing threatened hazel grouse populations in temperate forests.

1. Introduction

Nest survival is an important determinant of reproductive rates and, consequently, population dynamics. As the loss of nests is the main cause of reproductive failure in birds (Martin, 1995), nest survival also influences individual fitness, and females may select sites with particular characteristics that increase the probability that their clutches will hatch (Clark and Shutler, 1999). The structure of vegetation at or around nest sites is related to nest survival in many species of birds (Connelly et al., 1991; Riley et al., 1992), in part because it may help conceal nests, thereby affecting the probability of predation, typically the most important cause of nest loss (Ricklefs, 1969; Saniga, 2002). Attributes of nest locations have been shown to influence the nesting success of birds by affecting factors such as vulnerability to predation (Martin, 1993a; Kurki et al., 1998; Kauhala and Helle, 2002), brood parasitism (Burhans and Thompson, 1999), and resource acquisition (Huhta et al., 1999). Loss of nesting habitat and increases in predator population numbers can interact and be important causal factors in nest failures of ground-nesting birds (Martin, 1993b; Siepielski et al., 2001; Evans, 2004). Birds choose habitat with potentially higher fitness value for nest success (Newton, 1998).

The hazel grouse *Bonasa bonasia* is a Palaearctic species that is distributed over northern and central Eurasia (Johnsgard, 1983; Swenson et al., 1995), and is found in a wide variety of habitats from young successional to old-growth deciduous, mixed, and coniferous forests (Swenson and Danielson, 1991; Swenson and Angelstam, 1993; Rhim and Son, 2009). The southernmost edge of hazel grouse distribution in East Asia is located in the southern part of the Korean Peninsula (Rhim and Lee, 2001).

Also, this species is declining in numbers or has even gone extinct in many parts of its distribution range, primarily due to habitat disturbances by human activities (Rhim, 2010). Forest management is therefore considered to be important for the conservation and management of hazel grouse habitats. Little is known about the indirect or proximate effects of nest loss in hazel grouse generally or how vegetative characteristics of a temperate forest influence the nest survival of hazel grouse on the Korean Peninsula specifically. Forest managers and conservationists can benefit greatly from an understanding of the nest survival of hazel grouse to maintain the habitats and populations of this bird (Bergman et al., 1996).

My study focused on the nest survival of hazel grouse in a temperate forest in South Korea from 2003 to 2009. These hazel grouse are of conservational interest because the number of individuals and range of distribution is dramatically decreasing, due to habitat loss and fragmentation (Storch, 2000; Rhim and Lee, 2004). The first objective of this study was to estimate the true nest survival rates for radio-tagged hazel grouse. The second objective was to identify vegetation attributes that might influence nest success for management of forest as their habitat.

2. Methods

The study area was located in the experimental forest of the Gangwon Forest Development Institute in Chuncheon, Gangwon

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Province, Korea (37°48′ N, 127°48′ E). This 3000 ha study area is located near the southern edge of the hazel grouse distribution. The area consisted of several habitat types, most notably mixed forest (1000 ha), natural deciduous forest (800 ha), deciduous plantations (550 ha), coniferous plantations (450 ha), and rocky and bare habitats (200 ha). The dominant tree species in the mixed forest were Mongolian oak *Quercus mongolica* and Japanese red pine *Pinus densiflora*. Mongolian oak, Manchurian elm *Ulmus davidiana*, and Korean ash *Fraxinus rhynchophylla* were dominant in the natural deciduous forest. White birch *Betula plantyphylla*, Japanese larch *Larix leptolepis*, and Korean pine *Pinus koraiensis* dominated the deciduous and coniferous plantations. The forest types differed in respect to habitat characteristics, such as age, tree height, tree density, and tree dbh (diameter at breast height) (Table 1).

I captured female hazel grouse winter (December 2003–February 2009) by luring or chasing them into nylon fishing nets. After capture, hens were classified as either yearlings or adults on the basis of the shape and wearing of the outermost primary wing feathers (Bonczar and Swenson, 1992) and radio-marked with a 14-g necklace-type transmitter (Kenward, 1987; Millspaugh and Marzluff, 2001) that weighed 5% of the adult birds' mean weight. Individuals were located to within 20 m, five to ten times a week during the breeding season from March to June. Females were assumed to be nesting when movements became localized (Rhim, 2002). I found nests by tracking females approximately every other day, and by locating the female nesting in the understory vegetation using binoculars. The day on which nests were initiated was estimated by assuming a 25-day incubation period, 1.3 days/egg laid, and incubation commencing with the laying of the last egg (Johnsgard, 1983).

On the day of estimated hatching, the nest was revisited to assess its outcome. In most cases, the hatching date could be identified exactly, as the chicks were mostly still in, or very close to the nest. A nest was determined to be successful if $\geqslant 1$ chick had left the nest. For unsuccessful nests, the reason for lack of success (abandon/predation) was determined visually on the basis of eggshell remains and predator sign.

After the hen and the brood had departed from the nest, the habitat conditions were measured at each nest site ($n = 86$) within a circle of 5.56 m in diameter (0.01 ha). For each tree and snag within the circle, the species and the diameter at breast height (dbh) were recorded. The vertical layers within the circle were classified as understory (0–1 m), mid-story (1–2 m), sub-overstory (2–8 m), and overstory (8–20 m). Coverage was classified into the following four categories based on the percentage of cover of all vascular plants (trees, herbs, and shrubs) in each vertical layer as described by Rhim and Lee (1999): 0 (0%), 1 (1–33%), 2 (34–66%), 3 (67–100%). The visibility of the nest was further estimated by placing a grouse-sized object in the nest, and evaluating the distance from which the nest could be seen from the N, S, W, and E, from a height of approximately 1 m. Overall visibility was defined as the geometric mean of the visibility from the four directions.

Table 2
Number of hazel grouse nests used for analysis in a temperate forest, South Korea, during 2003–2009.

	Mixed forest	Natural deciduous forest	Deciduous plantation	Coniferous plantation	Total
2003	3	5	3	2	13
2004	4	8	2	0	14
2005	6	3	2	0	11
2006	4	6	3	0	13
2007	3	6	0	0	9
2008	5	8	3	2	18
2009	2	4	0	2	8
Total	27	40	13	6	86

The means of the variables used in the analyses of hazel gourse nest survival were compared between successful and unsuccessful nests using Mann–Whitney U tests. Nest survival models were used (Rotella et al. 2004) in Program MARK (White and Burnham, 1999) to estimate nest success and to test hypotheses about the relationships between vegetation variables and daily nest survival. For every mode, MARK produces the Akaike Information Criterion corrected for a small sample size (AIC_c), allowing models to be ranked according to the amount of information loss. Throughout the analysis, an information-theoretic philosophy of model selection was employed with a focus on multi-model inference (Burnham and Anderson, 2002), including habitat variables, such as tree density, basal area, coverage of overstory vegetation, coverage of sub-overstory vegetation, coverage of mid-story vegetation, coverage of understory vegetation, and visibility. The most parameterized model including covariates included year, forest type, and female age. Forest type was included in the analysis to control for large-scale spatial variation in nest survival, and female age was included to account for the potential that yearling females differed from older females in nest success. The daily survival rate was analyzed by repeated measures ANOVA with four forest stands for each of the seven years of the study. Duncan's multiple range test (DMRT) was used in the post hoc comparisons of mean values.

All possible models were assessed in order to identify the subset of models that were worse than and better than the threshold model. Akaike weights (ω) were determined for each variable present in at least one selected model. The average of parameter estimates (and confidence intervals) across the models where a given variable occurs (Burnham and Anderson, 2002) was computed, along with a 95% confidence interval, which allowed for

Table 1
Habitat characteristics of natural deciduous forest, mixed forest, deciduous plantation and coniferous plantation of the study area in a temperate forest, Chuncheon, Gangwon Province, South Korea.

	Mixed forest	Natural deciduous forest	Deciduous plantation	Coniferous plantation
Age (years)	52–67	57–68	30	43
Tree height (m)	24.2 ± 7.5[a]	25.3 ± 7.6	18.9 ± 5.2	20.6 ± 4.9
Tree density (no./ha)	211.5 ± 17.3	210.1 ± 13.7	189.2 ± 17.2	262.6 ± 28.8
Tree dbh[b] (cm)	22.8 ± 4.7	21.2 ± 7.8	21.9 ± 3.6	26.1 ± 3.8

[a] Mean ± SD.
[b] dbh: Diameter at breast height.

Table 3
Means (± standard errors) of the variables used in the analyses of hazel grouse nest survival for both successful ($N = 52$) and unsuccessful ($N = 34$) nests in a temperate forest, South Korea, during 2003–2009. *P* values are the results of the Mann–Whitney U tests.

Variables	Successful	Unsuccessful	*P*
Tree density (ind./ha)	245.42 ± 37.61	257.24 ± 42.09	0.26
Basal area (m^2/ha)	5.42 ± 1.78	5.24 ± 2.51	0.18
Coverage of overstory (8–20 m) vegetation	1.83 ± 0.34	1.53 ± 0.62	0.12
Coverage of sub-overstory (2–8 m) vegetation	1.34 ± 0.37	1.02 ± 0.53	0.10
Coverage of mid-story (1–2 m) vegetation	1.18 ± 0.21	1.42 ± 0.44	0.13
Coverage of understory (0–1 m) vegetation	2.72 ± 0.53	1.29 ± 0.32	0.01
Visibility	3.27 ± 0.26	5.91 ± 0.54	0.05

Table 4
Results of the hazel grouse nest survival models during 2003–2009 in a temperate forest, South Korea; AIC_c = Akaike Information Criterion, ΔAIC_c = Differences in AIC as compared with the best model, ω = Akaike weight. Analyses were performed using nest survival models in Program MARK.

Model	AIC_c	ΔAIC_c	ω	Deviance	No. of parameters
S(UC + FT + VS)	347.35	0.00	0.32	328.19	6
S(UC + FT)	348.87	1.52	0.19	330.26	5
S(FT + VS)	349.45	2.10	0.14	329.45	5
S(UC)	349.81	2.46	0.11	345.62	1
S(FT)	350.04	2.69	0.09	332.91	4
S(VS)	350.27	2.92	0.07	342.57	1
S(FT + TD + OC)	350.62	3.27	0.04	327.24	6
S(UC + AGE)	350.86	3.51	0.02	338.64	2
S(MC + TD)	350.86	3.51	0.02	335.21	2
S(OC + AGE)	351.09	3.74	0.00	337.43	2

AGE = age of the female at nesting, FT = forest types (mixed forest, natural deciduous forest, deciduous plantation, coniferous plantation), MC = mid-story coverage, OC = overstory coverage, SOC = sub-overstory coverage, TD = tree density, UC = understory coverage, VS = visibility.

the assessment of the extent of potential effects of covariates on daily nest survival.

3. Results

Seventy-one females were captured and nest survival was estimated for 86 nests from 2003 to 2009 (Table 2). Twenty-nine nests (33.7%) were depredated, five (5.8%) were abandoned, and 52 (60.5%) were successful. The estimated daily survival rate using maximum likelihood was 0.971 ± 0.005. The overall proportion of nesting hens was rather high throughout the study period, 85% on average. For the hens with two nests, the observed frequency of hens hatching zero, one, or both of their nests did not differ from expectations among the four forest types under a binomial distribution (χ^2 = 2.67, df = 6, P = 0.46). Eighty-six nests were located during egg laying (mean age = 8.2 days) and the results can be considered representative of the entire nesting period.

Table 5
Averaged Akaike weights (ω_{hat}), averaged parameter estimates (β_{hat}), and their standard errors (SE_{hat}), lower (LCI_{hat}) and upper (UCI_{hat}) confidence intervals.

	ω_{hat}	β_{hat}	SE_{hat}	LCI_{hat}	UCI_{hat}
Constant	1.000	1.569	0.398	0.367	4.215
Understory coverage	0.745	1.016	0.542	0.025	2.908
Forest type	0.628	0.562	0.354	0.184	0.923
Visibility	0.231	−0.086	0.024	−0.168	−0.024
Tree density	0.126	0.573	0.217	0.105	0.762
Overstory coverage	0.038	0.245	0.201	−0.459	0.839
Age	0.014	0.158	0.095	−0.026	0.467

The means and standard errors for all variables used in the models are shown in Table 3, for both successful and unsuccessful nests. Coverage of understory vegetation (Mann–Whitney U test, Z = 6.28, P = 0.01) and visibility (Z = 5.51, P = 0.01) differed significantly between successful and unsuccessful nests. The other measured habitat variables did not differ significantly (Z = −1.12–0.37, P = 0.10–0.26).

The best model of nest survival had an Akaike weight (ω) of 0.32, suggesting strong model uncertainty (Burnham and Anderson, 2002). The top-ranked model (r^2 = 0.76) explaining hazel grouse nest survival was 0.49 [understory coverage] + 0.31 [forest type] + 0.24 [visibility]. The second-ranked model (ω = 0.19) contained understory coverage and forest type for daily nest survival. The second-ranked model (r^2 = 0.59) was 0.37 [understory coverage] + 0.21 [forest type]. The third-ranked model (ω = 0.14, r^2 = 0.40) was 0.31 [forest type] + 0.18 [visibility] (Table 4). The top-ranked model indicated that there was variation in nest survival among stands, nest survival declined with visibility, and nest survival decreased with decreasing amounts of understory cover (Fig. 1). There was a general lack of support for models that contained other variables, such as tree density, basal area, mid-story coverage, overstory coverage or female age. The Akaike weights were 0.11, 0.09, and 0.07 for the fourth, fifth, and sixth-ranked models for nest survival, respectively.

Three predictor variables dominated: understory coverage, forest type, and visibility (Table 4). The sum of Akaike weights for

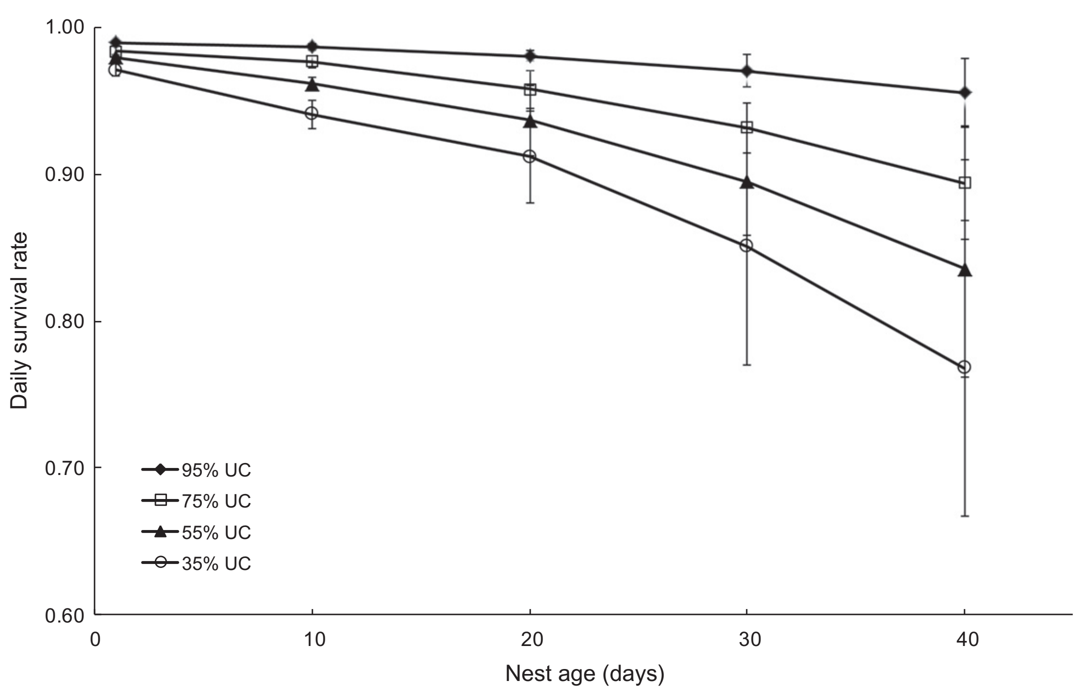

Fig. 1. Model-averaged daily survival rates and the 95% confidence intervals for hazel grouse nests in a temperate forest, South Korea, during 2003–2009 for varying amounts of understory coverage (UC).

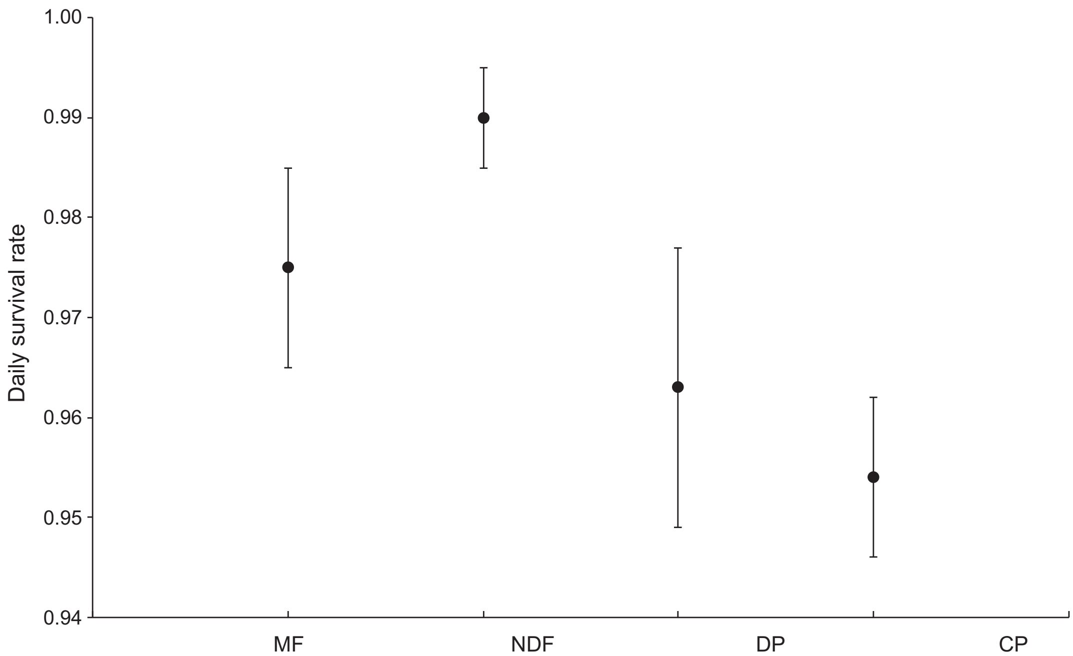

Fig. 2. Model-averaged daily survival rates and the 95% confidence intervals for hazel grouse nests in a temperate forest, South Korea, during 2003–2009. MF = mixed forest, NDF = natural deciduous forest, DP = deciduous plantation, CP = coniferous plantation.

models containing understory coverage was 0.745, with a model equal to the averaged β estimate for understory coverage of 1.016 ± 0.542. Daily survival decreased between 0.4% and 0.9% for every 20% decrease in understory coverage within 5.56 m of a nest site (Fig. 1). The sum of the Akaike weights for models containing forest stands was 0.628 (Table 5). Daily survival rates differed significantly among the forest stands (repeated measures ANOVA, $F_{3,18} = 14.39$, $P = 0.001$), with the highest found in natural deciduous forest (DMRT: $P = 0.05$) and the lowest in coniferous plantation (DMRT: $P = 0.05$) (Fig. 2). The sum of Akaike weights' for models containing visibility was 0.231, with a model equal to the averaged β estimate for understory coverage of -0.086 ± 0.024. The visibility had a negative regression coefficient (Table 5).

4. Discussion

The quality and amount of habitat on a larger spatial scale with estimation of survival rates, and variation in reproductive rates by habitat variables provide valuable insight into the influence of habitat constraints on the population dynamics of hazel grouse (Swenson, 1991; Rhim, 2010). The nest sites in the forest stands differed from that in the study area as a whole (Table 1), suggesting that the nest sites were not randomly located in the temperate forest.

Substantial variation was detected in daily nest survival among the different types of forest, with a trend of generally increasing nest success from the artificial stands (coniferous and deciduous plantations) to the natural stand (natural deciduous forest). Nest success in the natural deciduous forest was estimated to be about twice that of the coniferous plantation, 85% versus 41%. This pattern was a result of variation in habitat variables, because a natural deciduous forest has more dense understory coverage than planted stands (Rhim, 2006). The ground layer of natural deciduous and mixed forests might provide more food and cover for the birds (Sedgwick and Knopf, 1992). Rich forbs, buds, seeds, and catkins are foods, and dense understory cover serves as escape cover from predators (Rhim, 2006).

Natural selection should favor hazel grouse that choose nest sites to reduce the risk of predation, perhaps by providing concealment from potential predators (Martin, 1998) or modulating thermal flux (Deeming, 2002). The results showed hazel grouse nest success was significantly related to ground vegetation coverage. Therefore, from a habitat selection perspective, hens should select dense understory cover in natural deciduous forest or mixed forest and with low visibility. This is likely to be an adaptation to avian and mammalian predator behavior. High visibility of the nest and low understory cover possibly increase the detection radius, and/or increase the screening efficiency of the predator, and hence the probability of finding and depredating the nest (Storaas et al., 1999).

In this study, 60% of the nests succeeded and 34% of the nests were depredated. Potential hazel grouse egg and chick predators were corvid birds, particularly the jay *Garrulus glandarius*, carrion crow *Corvus corone*, and jungle crow *Corvus macrorhynchos*, Eurasian sparrowhawk *Accipiter nisus*, common kestrel *Falco tinnunculus*, Eurasian eagle owl *Bubo bubo*, and tawny owl *Strix aluco*. Among mammals, predators included the Siberian weasel *Mustela sibirica*, yellow-throated marten *Martes flavigula*, and Bengal cat *Felis bengalensis*. Small rodents can also be egg and chick predators. The main small rodent species are the Korean field mouse *Apodemus peninsulae*, Korean red-backed vole *Myodes regulus*, and striped field mouse *Apodemus agrarius*.

Maternal predictor variables were noticeably less important in mediating the risk of nest loss. The results did not reveal any differences in the daily nest survival rate between first breeders and older hens. Three other variables (overstory cover, mid-story cover, and basal area) were included in at least one of the best models. Their low average Akaike weight as well as confidence intervals including zero, however, suggest little, importance.

5. Conclusion

Nest success, by itself, may not be the principal determinant of breeding population size, because nest success may not be the only limiting factor for a given hazel grouse population. Other demographic rates, such as chick survival, juvenile survival, adult survival, natal dispersal, or movement, may be just as important to hazel grouse population dynamics. If any of these vital rates are under density-dependent control, the results of increased nest success may be counterbalanced by a reduction in another vital rate (e.g., chick survival). From an applied point of view, the results suggest that the two important factors for forest management, maintenance of understory cover and natural deciduous forest, are favorable for hazel grouse nest survival.

Nest success may also be influenced by other factors, such as the composition and behavior of the local predator community. Whether hazel grouse in our study were capable of perceiving predation risk, and adapting nest-site selection reproductive investments accordingly remains an open question and hence is a subject for further research.

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References

Bergman, H.H., Klaus, S., Müller, F., Scherzinger, W., Swenson, J.E., Wiesner, J., 1996. Die Haselhüner Bonasa bonasia und B. sewerzowi. Magdeburg, Wissenschaften.
Bonczar, Z., Swenson, J.E., 1992. Geographical variation in spotting patterns on hazel grouse *Bonasa bonasia* primary feathers: consequences for age determination. Ornis Fennica 69, 193–197.
Burhans, D.E., Thompson III, F.R., 1999. Habitat patch size and nesting success of yellow-breasted chats. Wilson Bulletin 111, 210–215.
Burnham, K.P., Anderson, D.R., 2002. Model Selection and Multimodel Inference. A Practical Information-Theoretic Approach. Springer, New York.
Clark, R.G., Shutler, D., 1999. Avian habitat selection: pattern from process in nest-use by ducks? Ecology 80, 272–287.
Connelly, J.W., Wakkinen, W.L., Apa, A.D., Reese, K.P., 1991. Sage grouse use of nest sites in southeastern Idaho. Journal of Wildlife Management 55, 521–524.
Deeming, D.C., 2002. Avian Incubation: Behavior, Environment, and Evolution. Oxford University Press, New York.
Evans, K.L., 2004. The potential for interactions between predator and habitat change to cause population declines of farmland birds. Ibis 146, 1–13.
Huhta, E., Jokimaki, J., Rahko, P., 1999. Breeding success of pied flycatchers in artificial forest edges: the effect of a sub-optimally shaped foraging area. Auk 116, 528–535.
Johnsgard, P.A., 1983. The Grouse of the World. University of Nebraska Press, Lincoln.
Kauhala, K., Helle, P., 2002. The impact of predator abundance on grouse populations in Finland: a study based on wildlife monitoring counts. Ornis Fennica 79, 14–25.
Kenward, R., 1987. Wildlife Radio Tagging: Equipment, Field Techniques and Data Analysis. Academic Press, London.
Kurki, S., Nikula, A., Helle, P., Lindén, H., 1998. Abundance of red fox and pine marten in relation to the composition of boreal forest landscapes. Journal of Animal Ecology 67, 874–886.
Martin, T.E., 1993a. Nest predation, nest sites, and birds: new perspectives on old patterns. Bioscience 43, 523–532.
Martin, T.E., 1993b. Nest predation among vegetation layers and habitat types: revising the dogmas. American Naturalist 14, 897–913.
Martin, T.E., 1995. Avian life history evolution in relation to nest sites, nest predation, and food. Ecological Monographs 65, 101–127.
Martin, T.E., 1998. Are microhabitat preferences of coexisting species under selection and adaptive? Ecology 79, 656–670.
Millspaugh, J.J., Marzluff, J.M., 2001. Radio Tracking and Animal Populations. Academic Press, San Diego.
Newton, I., 1998. Population Limitation in Birds. Academic Press, San Diego.
Rhim, S.J., 2002. Characteristics of distribution, behavior ecology and habitat use of hazel grouse Bonasa bonasia in Korea. PhD dissertation, Seoul National University, Seoul.
Rhim, S.J., 2006. Home range and habitat selection of hazel grouse *Bonasa bonasia* in a temperate forest of South Korea. Forest Ecology and Management 226, 22–25.
Rhim, S.J., 2010. Spring-season social organization of the hazel grouse (*Bonasa bonasia*) in relation to habitat type in temperate forests of South Korea. Ornis Fennica 87, 160–167.
Rhim, S.J., Lee, W.S., 1999. Differences in small mammal populations due to forest practices in deciduous forest. Journal of Korean Forest Society 88, 179–184.
Rhim, S.J., Lee, W.S., 2001. Characteristics of hazel grouse *Bonasa bonasia* distribution in southern Korea. Wildlife Biology 7, 257–261.
Rhim, S.J., Lee, W.S., 2004. Seasonal changes in territorial behavior of hazel grouse *Bonasa bonasia* in a temperate forest of South Korea. Journal of Ornithology 145, 31–34.
Rhim, S.J., Son, S.H., 2009. Natal dispersal of hazel grouse *Bonasa bonasia* in relation to habitat in a temperate forest of South Korea. Forest Ecological Management 258, 1055–1058.
Ricklefs, R.E., 1969. An analysis of nesting mortality in birds. Smithsonian Contributions to Zoology 9, 1–48.
Riley, T.Z., Davis, C.A., Ortiz, M., Wisdom, M.J., 1992. Vegetative characteristics of successful and unsuccessful nests of lesser prairie-chickens. Journal of Wildlife Management 56, 383–387.
Rotella, J.J., Dinsmore, S.J., Shaffer, T.L., 2004. Modeling nest-survival data: a comparison of recently developed methods that can be implemented in MARK and SAS. Animal Biodiversity and Conservation 27, 187–205.
Saniga, M., 2002. Nest loss and chick mortality in capercaillie (*Tetrao urogallus*) and hazel grouse (*Bonasa bonasia*) in West Carpathians. Felia Zoology 51, 205–214.
Sedgwick, J.A., Knopf, F.J., 1992. Describing willow flycatcher habitats: scales perspectives and gender differences. Condor 94, 720–733.
Siepielski, A.M., Rodewald, A.D., Yahner, R.H., 2001. Nest site selection and nesting success of the red-eyed vireo in central Pennsylvania. Wilson Bulletin 113, 302–307.
Storaas, T., Kastdalen, L., Wegge, P., 1999. Detection of forest grouse by mammalian predators: a possible explanation for high brood losses in fragmented landscapes. Wildlife Biology 5, 187–192.
Storch, I., 2000. Status survey and conservation action plan 2000–2004 Grouse. WPA/BirdLife/SSC Grouse Specialist Group, IUCN, Gland.
Swenson, J.E., 1991. Social organization of hazel grouse and ecological factors influencing it. PhD dissertation, University of Alberta, Alberta.
Swenson, J.E., Angelstam, P., 1993. Habitat separation by sympatric forest grouse in Fennoscandia in relation to boreal forest succession. Canadian Journal of Zoology 71, 1303–1310.
Swenson, J.E., Danielson, J., 1991. Status and conservation of hazel grouse in Europe. Ornis Scandinavian 22, 297–298.
Swenson, J.E., Andreev, A.V., Drovetskii, S.V., 1995. Factors shaping winter social organization in hazel grouse *Bonasa bonasia*: a comparative study in the eastern and western Palearctic. Journal of Avian biology 26, 4–12.
White, G.C., Burnham, K.P., 1999. Program MARK: survival estimation from populations of marked animals. Bird Study 46, 120–138.